

Learning in Matching

Shan Gui

Simin He

Gaoji Hu

Shanghai University of Finance and Economics

Wuhan Workshop on Experimental Economics

March 29, 2026

Motivation

- Two-sided matching is central to many real-world markets, including labor and marriage markets.
- Incomplete information:
 - firms may not know the productivity of their potential employees,
 - a man may have imprecise information about his girlfriend, etc.

match formation $\Leftrightarrow$ belief updating

- Fundamental questions to ask:
 - What is the theoretical prediction?
 - Will a “free market” settle as predicted?
 - If a market does not settle as predicted, why?

Literature

- Existing research has developed a rich theoretical framework under complete information [Gale and Shapley, 1962; Shapley and Shubik, 1971], and a few experimental studies have tested these predictions.
- Stability of matchings with incomplete information: Liu et al. [2014] and Liu [2020] Bikhchandani [2017], Chen and Hu [2020, 2023, 2024], Alston [2020], Pomatto [2022], Peralta [2025], Park [2025], Wang [2023], Hu [2025], Chen and Ho [2025], etc.
- Market dynamics:

	<i>complete information</i>	<i>incomplete information</i>
<i>non-strategic</i>	Roth and Vande Vate [1990] Rudov [2024], etc.	Chen and Hu [2020, 2024]
<i>strategic</i>	Pais [2008], Suh and Wen [2008] Haeringer and Wooders [2011]	Ferdowsian et al. [2025]

Literature

- Decentralized matching in experiments:

	<i>complete information</i>	<i>incomplete information</i>
<i>TU</i>	Otto and Bolle [2011] Dolgoplov et al. [2024]; He et al. [2024]	Nalbantian and Schotter [1995] Agranov et al. [2025]
<i>NTU</i>	Echenique et al. [2025]	Pais et al. [2020]

- All of them are testing complete-information stability concepts, the incomplete information is only implemented as an experimental friction!

Outline

Labor market: matching with one-sided incomplete information

1. Theoretical prediction:

stable matching with incomplete information.

(Bayesian, as well as prior-free)

2. Market dynamics (theoretical & experimental):

- decentralized matching markets with learning.
- learning revealed by matching.

3. Experimental results:

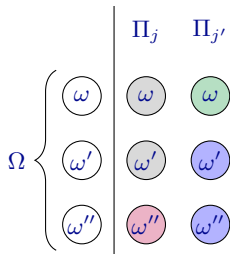
- whether stable matchings can be achieved. Learning $\Leftrightarrow$ Matching.
- the roles played by various *learning patterns*.

$$\Gamma = (\underbrace{I, J}_{\text{agents}}, \underbrace{\omega, \Omega, \beta}_{\text{info.}}, \underbrace{a, b}_{\text{payoffs}}): \text{ a matching market.}$$

- Agents
 - $I = \{1, \dots, n\}$: a finite set of workers.
 - $J = \{n+1, \dots, n+m\}$: a finite set of firms.
- Incomplete information (one-sided)
 - $\omega \in \Omega$: true state known by workers.
 - $\beta \in \Delta(\Omega)$: firms' prior belief.
- Matching values (one-to-one; no transfer)
 - $a_{ij}(\omega) \in \mathbb{R}$: for worker i .
 - $b_{ij}(\omega) \in \mathbb{R}$: for firm j .

Description of the market outcome

- A market *outcome* (μ, ω, Π) consists of
 1. a one-to-one matching $\mu : I \cup J \rightarrow I \cup J$, assumed to be publicly observable.
 2. a realized state ω , and
 3. an information structure for firms, denoted by $\Pi = (\Pi_{n+1}, \dots, \Pi_{n+m})$, where for each $j \in J$, Π_j is a partition over Ω .



Example: illustration of market outcomes

- $I = \{i_1, i_2\}$ and $J = \{j_1, j_2\}$.
- $\Omega = \{\omega, \omega', \omega''\}$ and $\beta(\omega) = \beta(\omega') = \beta(\omega'') = 1/3$.

		j_1		j_2	
i_1	ω	1	2	$-\infty$	$-\infty$
	ω'	1	4	$-\infty$	$-\infty$
	ω''	-1	-8	$-\infty$	$-\infty$
i_2	ω	$-\infty$	$-\infty$	1	2
	ω'	$-\infty$	$-\infty$	1	4
	ω''	$-\infty$	$-\infty$	1	-8

(A) (μ, ω, Π) : $\mu(i_1) = i_1$, $\mu(i_2) = i_2$; $\Pi_{j_1} = \{\{\omega, \omega', \omega''\}\}$, $\Pi_{j_2} = \{\{\omega, \omega', \omega''\}\}$.

(B) $(\hat{\mu}, \omega, \hat{\Pi})$: $\hat{\mu}(i_1) = j_1$, $\hat{\mu}(i_2) = i_2$; $\hat{\Pi}_{j_1} = \{\{\omega, \omega'\}, \{\omega''\}\}$, $\hat{\Pi}_{j_2} = \{\{\omega, \omega'\}, \{\omega''\}\}$.

Theoretical prediction: stable outcome

Definition (Liu, 2020; Chen and Hu, 2024)

An outcome (μ, ω, Π) is **stable** if it is **IR**, not **blocked**, and “**information stable**”.

1. An outcome (μ, ω, Π) is **individually rational (IR)** if

$$a_{i\mu(i)}(\omega) \geq 0 \text{ for all } i \in I \quad \text{and}$$

$$\mathbb{E} [b_{\mu(j)j} | \Pi_j(\omega)] \geq 0 \text{ for all } j \in J.$$

2. An outcome (μ, ω, Π) is **blocked** by (i, j) if

$$a_{ij}(\omega) > a_{i\mu(i)}(\omega) \quad \text{and}$$

$$\mathbb{E} [b_{ij} | \Pi_j(\omega) \cap D_{\mu, ij}] > \mathbb{E} [b_{\mu(j)j} | \Pi_j(\omega) \cap D_{\mu, ij}].$$

3. An outcome (μ, ω, Π) satisfies **information stability** if $\Pi \vee \mathcal{N}_{\mu, \Pi} = \Pi$.

Notations: $N_{\mu, \Pi} := \{\tilde{\omega} \in \Omega : (\mu, \tilde{\omega}, \Pi) \text{ is IR and not blocked}\}$; $\mathcal{N}_{\mu, \Pi} := \{N_{\mu, \Pi}, \Omega \setminus N_{\mu, \Pi}\}$.

Example: stable outcome and learning dynamics

		j_1		j_2	
i_1	ω	1	2	$-\infty$	$-\infty$
	ω'	1	4	$-\infty$	$-\infty$
	ω''	-1	-8	$-\infty$	$-\infty$
i_2	ω	$-\infty$	$-\infty$	1	2
	ω'	$-\infty$	$-\infty$	1	4
	ω''	$-\infty$	$-\infty$	1	-8

(A) (μ, ω, Π) : $\mu(i_1) = i_1$, $\mu(i_2) = i_2$; $\Pi_{j_1} = \{\{\omega, \omega', \omega''\}\}$, $\Pi_{j_2} = \{\{\omega, \omega', \omega''\}\}$.

(B) $(\hat{\mu}, \omega, \hat{\Pi})$: $\hat{\mu}(i_1) = j_1$, $\hat{\mu}(i_2) = i_2$; $\hat{\Pi}_{j_1} = \{\{\omega, \omega'\}, \{\omega''\}\}$, $\hat{\Pi}_{j_2} = \{\{\omega, \omega'\}, \{\omega''\}\}$.

The stable outcome:

(C) $(\check{\mu}, \omega, \check{\Pi})$: $\check{\mu}(i_1) = j_1$, $\check{\mu}(i_2) = j_2$; $\check{\Pi}_{j_1} = \{\{\omega, \omega'\}, \{\omega''\}\}$, $\check{\Pi}_{j_2} = \{\{\omega, \omega'\}, \{\omega''\}\}$.

(A) $\xrightarrow{\text{learning from conditional evaluation (CE)}} (B) \xrightarrow{\text{learning from blocking (LB)}} (C)$

Partial revealing: firms cannot pin down the true state.

Example 2: stable outcome and learning dynamics

		j_1		j_2	
i_1	ω	-1	2	$-\infty$	$-\infty$
	ω'	1	4	$-\infty$	$-\infty$
	ω''	1	8	$-\infty$	$-\infty$
i_2	ω	$-\infty$	$-\infty$	1	2
	ω'	$-\infty$	$-\infty$	1	-4
	ω''	$-\infty$	$-\infty$	1	-8

(X) (μ, ω, Π) : $\mu(i_1) = i_1$, $\mu(i_2) = i_2$; $\Pi_{j_1} = \{\{\omega, \omega', \omega''\}\}$, $\Pi_{j_2} = \{\{\omega, \omega', \omega''\}\}$.

(W) $(\bar{\mu}, \omega, \bar{\Pi})$: $\bar{\mu}(i_1) = i_1$, $\bar{\mu}(i_2) = j_2$; $\bar{\Pi}_{j_1} = \{\{\omega\}, \{\omega', \omega''\}\}$, $\bar{\Pi}_{j_2} = \{\{\omega\}, \{\omega', \omega''\}\}$.

(X) $\xrightarrow{\text{learning from no blocking (LNB)}}$ (W)

Full revealing: firms can pin down to the true state.

Learning patterns

	Fully revealing	Partially revealing
Learning from conditional evaluation	CE-Full	CE-Partial
Learning from blocking	LB-Full	LB-Partial
Learning from no blocking	LNB-Full	LNB-Partial
Compound learning	CE+LB	
	CE+LNB	

- How to identify each learning pattern in some market?
 1. unique stable matching.
 2. unique “learning-and-blocking” path from autarky to stability.
 3. each *learning pattern* can be uniquely identified by *evolution of matches*.

Experiment design: treatment variables

- Main market type: **learning types**.
 - Plain, CE, LB, LNB, Compound.
- Secondary market type (subtype): **revelation level**.
 - **Full** revelation: can uniquely pin down the true state
 - **Partial** revelation: can only rule out one state, but two states remain possible
- Within-subject design: each subject experiences all types of markets.

Experiment design: markets and order

- There are 20 markets in total: 5 learning types $\times$ 2 subtypes $\times$ 2 parameter sets.
- Each subject experiences all 20 markets.
- We use a Latin-Square method to control for order effect.

Block A	Block B	Block C	Block D
Plain	Plain	Plain	Plain
CE-Full	LB-Partial	LNB-Full	Comp: CE+LNB
LB-Full	LNB-Partial	Comp: CE+LB	CE-Partial
LNB-Full	Comp: CE+LNB	CE-Full	LB-Partial
Comp: CE+LB	CE -Partial	LB-Full	LNB-Partial

Treatment order 1:	A	B	C	D
Treatment order 2:	B	C	D	A
Treatment order 3:	C	D	A	B
Treatment order 4:	D	A	B	C

Experiment design: roles and random-matching

- 192 subjects in total, equally distributed across the four treatment orders.
- Each large *matching group* consists of 12 participants.
 - 6 workers + 6 firms (roles fixed).
- In each round, 6 workers + 6 firms are randomly matched to form three matching markets, each consisting of 2 workers and 2 firms.
- The 20 markets are played in 20 rounds.

Experiment design: pre-randomization

- We perform pre-randomization to ensure that:
 - each firm-role subject experiences 8 easier roles and 8 harder roles (except the 4 markets in Plain, where there exist no easy or hard roles).
 - each state $\omega, \omega', \omega''$ is independently drawn across 20 markets.
 - 4 payoff matrices for each pattern: original, row swap, column swap, both swap.

Experiment design: how to form a match

Experiment Round No. 2

Timer: 27s

Only Lemon and Mango know the weather.

Matching Payoff Table

		Yellow		Green		
Lemon		2	-6	6	5	Sunny
		-4	5	9	8	Cloudy
		-7	1	3	2	Rainy
Mango		5	1	-1	2	Sunny
		-3	-9	-3	-9	Cloudy
		8	6	-7	-4	Rainy

My Role: Green

Weather in this Round: Unknown for me

Sending Invitations

Please choose the Fruit you would like to invite:

Receiving Invitations

Mango sent you an invitation.

Please response within 11s:

Current "Temporary Matches"

Highlighted cells in the Payoff Table show the current matches:
Mango is matched with Yellow, Lemon is matched with Green.

History "Temporary Matches"

At 40s, Lemon was matched with Green.
At 32s, Mango was matched with Yellow.
At 20s, Lemon and Yellow were unmatched.
At 10s, Mango and Green were unmatched.

My Current "Temporary Match"

Matched with Lemon. Click the button if you would
like to dissolve the match:

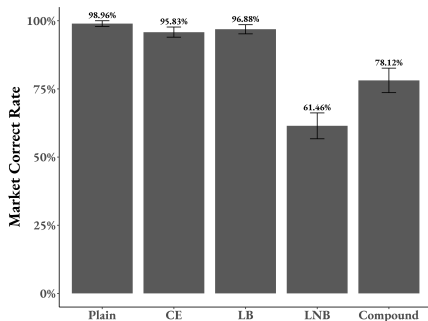
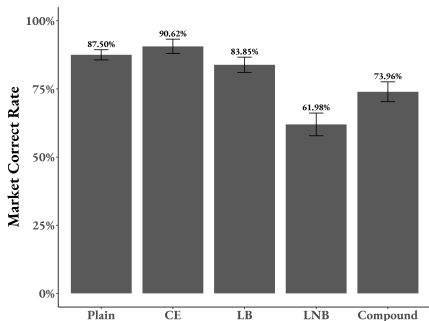
My Activity History

At 69s, Mango sent me an invitation.
At 57s, I rejected Mango's invitation.
At 51s, Mango sent me an invitation.
At 40s, Lemon accepted my invitation and

Experiment design: decentralized matching protocol

- Workers and firms may freely send proposals to one opponent at a time, and may leave a match freely.
- Each market runs for at least 60 seconds.
- A public information area displays a countdown timer above the payoff matrix.
- When there is a change in the temporary matches:
 - the 60-second timer restarts
 - the corresponding payoff cell lights up or dims
- The market ends when the timer reaches zero (i.e., the market is inactive for 60 seconds), or when they all decide to proceed to the next round, and then all matches become permanent.

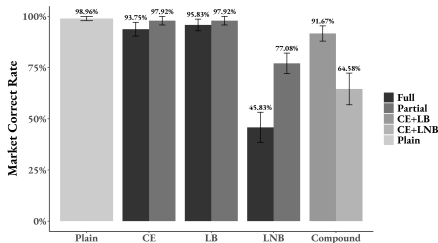
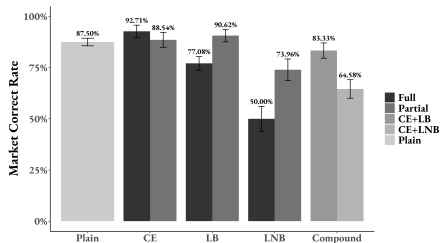
Result 1: market-level correctness rates (all-rounds and last 10 rounds)



Result (market-level correctness rates)

- All 20 rounds: $\text{Plain} = \text{CE} >^{**} \text{LB} >^{**} \text{Compound} >^{**} \text{LNB}$.
- Last 10 rounds: $\text{Plain} = \text{CE} = \text{LB} >^{***} \text{Compound} >^{**} \text{LNB}$.

Result 2: market-level correctness rates by subtypes



Result (market-level correct rates by subtypes)

All 20 rounds

$CE\text{-}Partial = CE\text{-}Full$

$LB\text{-}Partial >^{***} LB\text{-}Full$

$LNB\text{-}Partial >^{**} LNB\text{-}Full$

$CE+LB >^{***} CE+LNB$

Last 10 rounds

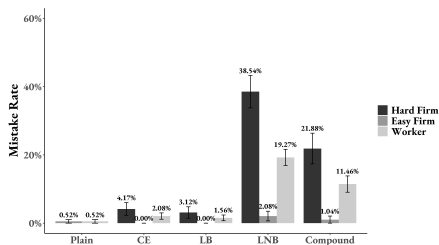
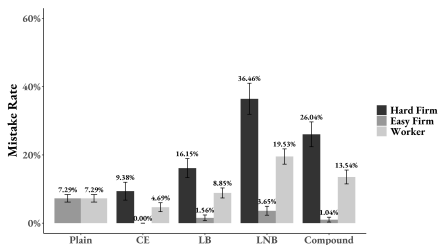
$CE\text{-}Partial = CE\text{-}Full$

$LB\text{-}Partial = LB\text{-}Full$

$LNB\text{-}Partial >^{***} LNB\text{-}Full$

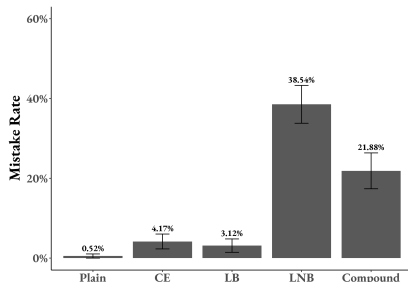
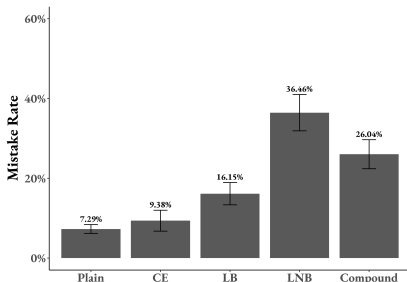
$CE+LB >^{***} CE+LNB$

Where do those *mistakes* come from?



- Workers make passive mistakes; so ignore them.
- “Hard” firms account for most *mistakes*! — Learning is crucial!

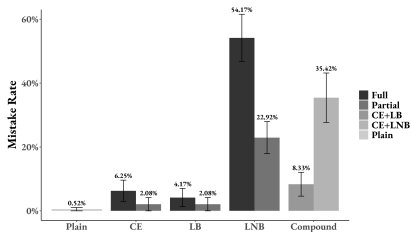
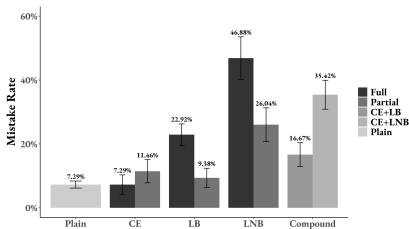
Result 3: individual-level: hard-firm mistake rates



Result (hard-firm mistake rates)

- All 20 rounds: $\text{Plain} = \text{CE} <^{**} \text{LB} <^{**} \text{Compound} \leq^* \text{LNB}$.
- Last 10 rounds: $\text{Plain} = \text{CE} = \text{LB} <^{***} \text{Compound} <^{**} \text{LNB}$. (3-step reasoning)

Result 4: individual-level: hard-firm mistake rates by subtypes



Result (hard-firm mistake rates by subtypes)

All 20 rounds

$LB\text{-}Partial <^{***} LB\text{-}Full$

$LNB\text{-}Partial <^{**} LNB\text{-}Full$

$CE+LB <^{***} CE+LNB$

Last 10 rounds

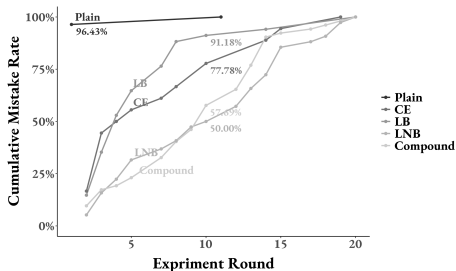
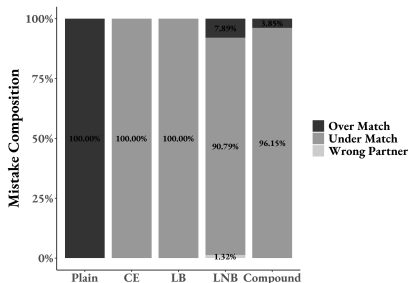
$LB\text{-}Partial = LB\text{-}Full$

$LNB\text{-}Partial <^{***} LNB\text{-}Full$

$CE+LB <^{***} CE+LNB$

(loss aversion)

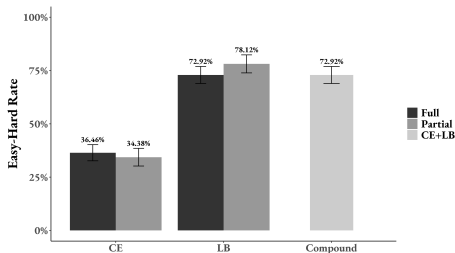
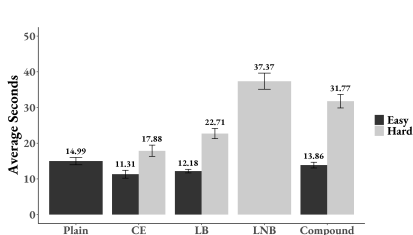
Result 5: mistake types and evolution



Result (mistake types)

<i>Over-match</i>	<i>Plain</i>		<i>LNB</i>	<i>Compound</i>
<i>Under-match</i>		<i>CE, LB</i>	<i>LNB</i>	<i>Compound</i>
<i>Wrong-partner</i>			<i>LNB</i>	
	<i>mostly round 1!</i>	<i>mostly rounds 2-10</i>	<i>evenly distributed</i>	

Result 6: matching dynamics: easy vs. hard matches



Result (matching dynamics)

- *Hard matches require substantially more time than easy matches.*

time required for hard-match: $LNB > Compound > LB > CE$

- *Ordinal “easy before hard”: LB and $CE+LB$ more often than CE .*

Take-away messages

CE:	CE-Full	CE-Partial
LB:	LB-Full	LB-Partial
LNB:	LNB-Full	LNB-Partial
Compound:	CE+LB	CE+LNB

1. Market-level correctness rates:

- Plain = CE > LB > Compound > LNB: **3-step reasoning in LNB**.
- *Partial* > *Full* in LB and LNB: **loss aversion**

2. “Hard” firms account for most mistakes. —Learning is crucial!

- *Over-match* in Plain, *Under-match* in CE and LB: mistakes **vanish**.
- *O + U* in Compound, *O + U + Wrong-partner* in LNB: mistakes **persist**.

3. Market dynamics:

- Time required: LNB 37'' > Compound 32'' > LB 23'' > CE 18'' > Plain 15''.
- Ordinal “easy before hard”: more often in LB and CE+LB than in CE.

- **Learning from no blocking** is very challenging, how can this experimental finding inform the theory?
- Alternative decentralized implementations of stability notions:
 - negotiation or communication instead of proposing and rejecting/accepting
 - testing structured dynamic games to examine whether predicted strategies are adopted
- Alternative stability notions in theory?

References

- AGRANOV, M., A. DIANAT, L. SAMUELSON, AND L. YARIV (2025): “Paying to Match: Decentralized Markets with Information Frictions,” *forthcoming at The Review of Economic Studies*.
- ALSTON, M. (2020): “On the non-existence of stable matches with incomplete information,” *Games and Economic Behavior*, 120, 336–344.
- BIKHCHANDANI, S. (2017): “Stability with One-Sided Incomplete Information,” *Journal of Economic Theory*, 168, 372–399.
- CHEN, Y.-C. AND S. C. S. HO (2025): “Efficiency of Stable Matchings with Two-Sided Incomplete Information,” *Available at SSRN: <https://ssrn.com/abstract=3496454>*.
- CHEN, Y.-C. AND G. HU (2020): “Learning by matching,” *Theoretical Economics*, 15, 29–56.
- (2023): “A theory of stability in matching with incomplete information,” *American Economic Journal: Microeconomics*, 15, 288–322.
- (2024): “Bayesian stable states,” *Games and Economic Behavior*, 145, 102–116.
- DOLGOPOLOV, A., D. HOUSER, C. MARTINELLI, AND T. STRATMANN (2024): “Assignment markets: Theory and experiments,” *European Economic Review*, 165, 104738.
- ECHENIQUE, F., A. ROBINSON-CORTÉS, AND L. YARIV (2025): “An experimental study of decentralized matching,” *Quantitative Economics*, 16, 497–533.
- FERDOWSIAN, A., M. NIEDERLE, AND L. YARIV (2025): “Strategic decentralized matching: The effects of information frictions,” *Princeton and Stanford Universities*.
- GALE, D. AND L. S. SHAPLEY (1962): “College admissions and the stability of marriage,” *The American Mathematical Monthly*, 69, 9–15.
- HAERINGER, G. AND M. WOODERS (2011): “Decentralized job matching,” *International Journal of Game Theory*, 40, 1–28.
- HE, S., J. WU, H. ZHANG, AND X. ZHU (2024): “Decentralized matching with transfers: experimental and noncooperative analyses,” *American Economic Journal: Microeconomics*, 16, 406–439.
- HU, G. (2025): “The Structure of Bayesian Stable Matchings,” *Available at SSRN 4910284*.
- LIU, Q. (2020): “Stability and Bayesian Consistency in Two-Sided Markets,” *American Economic Review*, 110, 2625–2666.
- LIU, Q., G. J. MAILATH, A. POSTLEWATE, AND L. SAMUELSON (2014): “Stable Matching with Incomplete Information,” *Econometrica*, 82, 541–587.
- NALBANTIAN, H. R. AND A. SCHOTTER (1995): “Matching and Efficiency in the Baseball Free-Agent System: An Experimental Examination,” *Journal of Labor Economics*, 13, 1–31.